

Weight Calculation & Promotion Logic Reference

This document details the dynamic, real-time shipment weight calculation system and promotional mechanisms utilized in the **Layo Shipping Booking & Estimation Portal**. The implementation provides accurate weight aggregation, dynamic promotional discounting, and resilient client-side estimation.

1. Standard Item Weights Map

Every item registered in Layo has a baseline unit weight (in kg) based on its category and subcategory:

Category	Subcategory	Standard Weight
Clothing	Light Topwear (T-shirts, Kurtis, etc.)	0.20 kg (200g)
Clothing	Light Bottoms (Shorts, Leggings)	0.25 kg (250g)
Clothing	Heavy Bottoms (Jeans, Trousers)	0.50 kg (500g)
Clothing	Light Dresses & Sets	0.40 kg (400g)
Clothing	Heavy Outerwear (Jackets, Sweaters)	0.90 kg (900g)
Clothing	Heavy Ethnic & Party (Sarees, Lehengas)	1.00 kg (1000g)
Clothing	Heavy Winter Sets	1.30 kg (1300g)
Clothing	Small Accessories (Socks, Ties, Innerwear)	0.05 kg (50g)*
Footwear	Sandals / Flip-flops	0.40 kg (400g)
Footwear	Casual Shoes / Sneakers	0.80 kg (800g)
Footwear	Formal Shoes	1.00 kg (1000g)
Footwear	Boots	1.50 kg (1500g)
Jewelry	Earrings / Rings	0.05 kg (50g)
Jewelry	Necklaces / Bangles	0.20 kg (200g)
Jewelry	Heavy Sets	0.50 kg (500g)
Home Decor	Cushion Covers / Linens	0.30 kg (300g)
Home Decor	Artifacts (Small)	0.50 kg (500g)
Home Decor	Wall Hangings	0.80 kg (800g)

2. '5 Free Essentials' Promotional Math

3. Implementation Code

```
const totals = useMemo(() => {
  let totalWeight = 0;
  let smallAccessoriesCount = 0;

  items.forEach(item => {
    let unitWeight = shippingData[item.category]?.[item.subcategory] || 0.2;

    if (item.subcategory.includes("Small Accessories")) {
      let itemWeight = 0;
      for (let q = 0; q < item.quantity; q++) {
        smallAccessoriesCount++;
        if (smallAccessoriesCount > 5) {
          itemWeight += 0.05; // 50g standard weight
        } else {
          itemWeight += 0;    // 0g promotional weight
        }
      }
      totalWeight += itemWeight;
    } else {
      totalWeight += unitWeight * item.quantity;
    }
  });
  return { weight: totalWeight.toFixed(2), ... };
}, [items]);
```

4. Scenario Dry Run